

PrimeCore Minerals

VLAN 99

Management

Network

Implementation

Prepared by: Jared Williams
Date: September 29th, 2026
Role: Security and Validation Lead



1. Introduction.....	2
2. Purpose and Scope.....	2
3. Management VLAN Design.....	2
4. Brisbane HQ Implementation.....	3
4.1 Management Addresses.....	3
4.2 Open vSwitch Management Interface.....	4
4.3 HQ Firewall and Management Access Controls.....	5
5. Mine Site A Implementation.....	7
5.1 Management Addressing.....	7
5.2 Site A Firewall Controls.....	7
5.3 Site A Switch Management.....	9
6. Testing and Validation.....	10
7. Limitations and Future Improvements.....	11
8. Conclusion.....	11

1. Introduction

This document provides an explanation of how the VLAN 99 Management network was implemented across the PrimeCore Minerals Brisbane HQ and Mine Site A networks. VLAN 99 was created to keep management traffic separate from normal employee and visitor traffic while providing a separate network for managing devices such as pfSense and the Open vSwitch.

2. Purpose and Scope

The Management VLAN was created to keep network administration separate from general employee and Visitor traffic. Instead of allowing regular user networks to access devices such as pfSense, Open vSwitch and other management services, VLAN 99 provides a separate network for authorised management devices.

- Keep management traffic separate from Employee and Visitor networks.
- Give pfSense and network switches their own management addresses.
- Block Employee and Employee Wi-Fi devices from accessing the Management VLAN.
- Limit management access to approved management workstations instead of allowing the whole VLAN unrestricted access.
- Provide secure access to services such as the HQ Grafana dashboard.
- Test that authorised management access works while normal user access remains blocked.

3. Management VLAN Design

Using VLAN 99 at both locations keeps the logical design consistent. The different 192.168.99.0/24 and 192.168.199.0/24 subnets prevent address overlap and allow each site to apply its own local firewall and management controls.

Location	VLAN	Network	Gateway	Authorised Management PC
Brisbane HQ	99	192.168.99.0/24	192.168.99.1	HQ-MGMT-PC - 192.168.99.71
Mine Site A	99	192.168.199.0/24	192.168.199.1	SITE-A-MGMT-PC - 192.168.199.71

4. Brisbane HQ Implementation

The management PC “HQ-MGMT-PC” was added to VLAN 99 to access both pfSense and the HQ Open vSwitch. The Management VLAN uses the 192.168.99.0/24 network and was set up to keep management traffic separate from normal user traffic at Brisbane HQ. Devices that are used to manage the network are placed on VLAN 99 so they can access important network devices without allowing Employee and Visitor networks to access the same management interfaces.

4.1 Management Addresses

Device	Management IP Address	Purpose
HQ-FW (pfSense)	192.168.99.1	Management VLAN gateway and firewall administration
HQ Open vSwitch (HQ-SW-OVS)	192.168.99.60	Open vSwitch management interface
HQ Management PC (HQ-MGMT-PC)	192.168.99.71	Authorised management workstation

This setup allows the HQ Management PC to access and manage pfSense and the Open vSwitch through VLAN 99 instead of Employee or Visitor networks. It also makes the management traffic easier to control with firewall rules and restricts access to pfSense and the Open vSwitch.

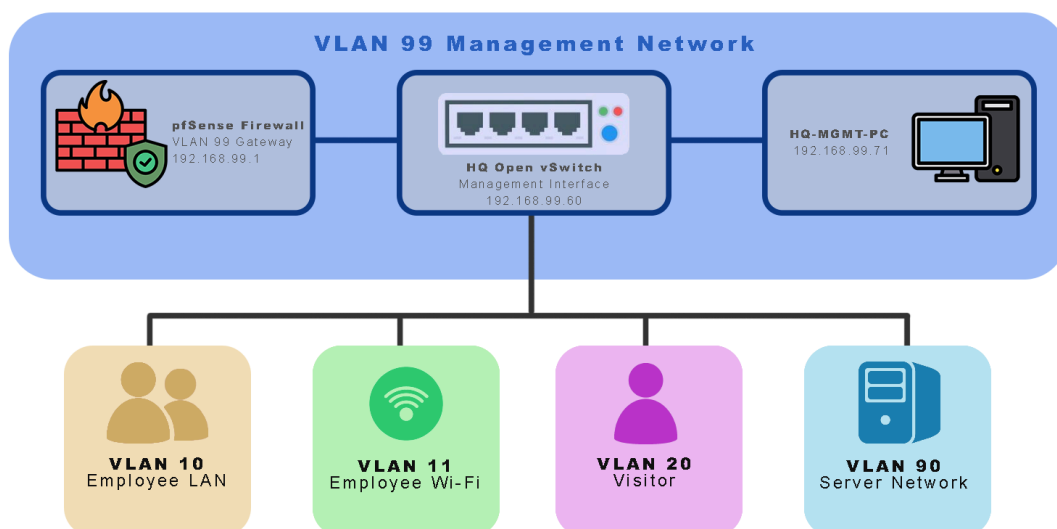


Figure 1: VLAN 99 management network design at Brisbane HQ

4.2 Open vSwitch Management Interface

An internal interface called mgmt99 was created to make the Open vSwitch accessible from VLAN 99. The interface was assigned to VLAN 99 and given the static IP address 192.168.99.60/24.

```
sudo ovs-vsctl --may-exist add-port br0 mgmt99
sudo ovs-vsctl set interface mgmt99 type=internal
sudo ovs-vsctl set port mgmt99 tag=99
sudo ip addr replace 192.168.99.60/24 dev mgmt99
```

These commands create the mgmt99 interface, assign it to VLAN 99 and configure the management address 192.168.99.60/24. This allows the Open vSwitch to be managed through the dedicated Management VLAN.

```
Bridge br0
  Port ens10
    tag: 99
    Interface ens10
  Port ens7
    tag: 10
    Interface ens7
  Port ens14
    tag: 91
    Interface ens14
  Port ens12
    Interface ens12
  Port mgmt99
    tag: 99
    Interface mgmt99
      type: internal
  Port ens5
    tag: 30
    Interface ens5
  Port ens6
    tag: 50
    Interface ens6
  Port ens9
    tag: 11
    Interface ens9
  Port ens4
    trunks: [10, 11, 20, 30, 40, 41, 50, 90, 91, 99]
    Interface ens4
  Port ens11
    tag: 90
    Interface ens11
  Port ens8
    tag: 20
    Interface ens8
  Port ens3
    tag: 99
    trunks: [11, 20]
    Interface ens3
  Port br0
    Interface br0
      type: internal
  Port ens13
    tag: 90
    Interface ens13
  ovs_version: "3.7.1"
labadmin@HQ-SW-OVS:~$ ip -br addr show mgmt99
mgmt99      UNKNOWN      192.168.99.60/24  fe80::60ca:8ff:feea:5e0c/64
labadmin@HQ-SW-OVS:~$
```

Figure 2: HQ Open vSwitch output showing the mgmt99 interface on VLAN 99 with 192.168.99.60/24.

4.3 HQ Firewall and Management Access Controls

The firewall controls were used to keep the Management VLAN separated from normal user networks while still allowing the authorised management workstation to reach required administrative services.

Control	Source	Destination	Service / Result
Employee management restriction	HQ Employee VLAN	HQ Management VLAN	Blocked and logged
Employee Wi-Fi management restriction	HQ Employee Wi-Fi VLAN	HQ Management VLAN	Blocked and logged
pfSense management	HQ-MGMT-PC 192.168.99.71	HQ-FW 192.168.99.1	Management access permitted
Grafana monitoring access	HQ-MGMT-PC 192.168.99.71	HQ-MONITOR 192.168.90.60	TCP 3000 allowed

VLAN 99 is also used to control access to the Grafana monitoring dashboard. HQ-MGMT-PC on VLAN 99 is allowed to connect to HQ-MONITOR on TCP port 3000, while normal Employee and Visitor networks are not given access to the dashboard.

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	0/0 B	IPv4 *	EMPLOYEE_VLAN subnets	*	MANAGEMENT_NETWORK	*	*	none		R05 - Block Employee Access to Management	
☐	0/5 KIB	IPv4 ICMP any	EMPLOYEE_VLAN subnets	*	192.168.110.0/24	*	*	none		allow HQ employee icmp to site a	
☐	0/261.16 MiB	IPv4 TCP	EMPLOYEE_VLAN subnets	*	*	WEB_PORTS	*	none		R02a - Allow Employee Web Access	
☐	0/0 B	IPv4 TCP	EMPLOYEE_VLAN subnets	*	PRINTER_NETWORK	PRINT_PORTS	*	none		R09a - Allow Employee LAN to Printers	
☐	0/2.80 MiB	IPv4 TCP/UDP	EMPLOYEE_VLAN subnets	*	*	DNS_PORT	*	none		R02b - Allow Employee DNS	
☐	0/0 B	IPv4 TCP	EMPLOYEE_LAN	*	AD_SERVER	AD_TCP_PORTS	*	none		Allow employee lan to active directory tcp	
☐	0/0 B	IPv4 UDP	EMPLOYEE_LAN	*	AD_SERVER	AD_UDP_PORTS	*	none		Allow employee lan to active directory udp	

Figure 3a: HQ Employee VLAN rule blocking access to the Management VLAN.

Rules (Drag to Change Order)											
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
	0/0 B	IPv4 *	EMPLOYEE_WIFI_VLAN subnets	*	MANAGEMENT_NETWORK	*	*	none		R06 - Block Employee WIFI Access to Management	
	0/356 B	IPv4 ICMP any	EMPLOYEE_WIFI_VLAN subnets	*	*	*	*	none		TEMP - Employee wifi connection	
	0/672.16 MiB	IPv4 TCP	EMPLOYEE_WIFI_VLAN subnets	*	*	WEB_PORTS	*	none		R03a - Allow Employee WIFI Web Access	
	0/0 B	IPv4 TCP	EMPLOYEE_WIFI_VLAN subnets	*	PRINTER_NETWORK	PRINT_PORTS	*	none		R09b - Allow Employee WIFI for Printers	
	0/11.91 MiB	IPv4 TCP/UDP	EMPLOYEE_WIFI_VLAN subnets	*	*	DNS_PORT	*	none		R03b - Allow Employee WIFI DNS	
	1/17.04 MiB	IPv4 TCP	EMPLOYEE_WIFI	*	AD_SERVER	AD_TCP_PORTS	*	none			
	0/0 B	IPv4 UDP	EMPLOYEE_WIFI	*	AD_SERVER	AD_UDP_PORTS	*	none			

Figure 3b: HQ Employee Wi-Fi rule blocking access to the Management VLAN.

Rules (Drag to Change Order)											
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
	3/58.87 MiB	IPv4 TCP	MANAGEMENT_VLAN subnets	*	This Firewall (self)	MANAGEMENT_PORTS	*	none		R10b -	
	0/0 B	IPv4 TCP	MANAGEMENT_VLAN subnets	*	This Firewall (self)	MANAGEMENT_PORTS	*	none		R10a - Allow Management Access to pfSense	
	0/2.78 MiB	IPv4 TCP/UDP	HQ_AP	*	*	DNS_PORT	*	none		HQ AP DNS	
	0/263.20 MiB	IPv4 TCP	HQ_AP	*	*	WEB_PORTS	*	none			
	0/0 B	IPv4 UDP	*	*	*	NTP_PORT	*	none			
	2/9.41 MiB	IPv4 TCP	192.168.99.71	*	192.168.90.60	3000 (HBCI)	*	none			

Figure 3c: HQ Management VLAN rules allowing authorised management access to pfSense and HQ-MGMT-PC access to Grafana on TCP port 3000.

5. Mine Site A Implementation

5.1 Management Addressing

The Management VLAN was also implemented at Mine Site A so the site could use the same management setup as Brisbane HQ. VLAN 99 uses the 192.168.199.0/24 network at Site A and is used for managing important devices such as the pfSense firewall and Site A switch.

Device	Management IP Address	Purpose
SITE-A-FW (pfSense)	192.168.199.1	Management VLAN gateway and firewall administration
PCM-SITEA-SW01	192.168.199.10	Site A switch management
SITE-A-MGMT-PC	192.168.199.71	Authorised Site A management workstation

5.2 Site A Firewall Controls

These controls make SITE-A-MGMT-PC the main device used to manage the Site A network, while Employee and Employee Wi-Fi devices remain blocked from accessing the Management VLAN.

Control	Source	Destination	Service	Action
Management DNS	192.168.199.71	192.168.199.1	TCP/UDP 53	Allow
Gateway ICMP	192.168.199.71	192.168.199.1	ICMP	Allow
Admin to internal networks	192.168.199.71	SITEA_INTERNAL_NETWORKS	Any	Allow
Management web access	192.168.199.71	Any	TCP 80/443	Allow
Employee management restriction	Site A Employee VLAN 10	Site A Management VLAN 99	Any	Block + Log
Employee Wi-Fi management restriction	Site A Employee Wi-Fi VLAN 11	Site A Management VLAN 99	Any	Block + Log

Rules (Drag to Change Order)											
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
	✓ 1/222 KIB	*	*	*	SITEA_MGMT Address	443 80	*	*		Anti-Lockout Rule	
	✓ 4/8.93 MiB	IPv4 TCP/UDP	192.168.199.71	*	192.168.199.1	53 (DNS)	*	none		R01 - Management DNS	
	✓ 0/0 B	IPv4 ICMP any	192.168.199.71	*	192.168.199.1	*	*	none		R02 - Management Gateway ICMP	
	✓ 0/0 B	IPv4 *	192.168.199.71	*	SITEA_INTERNAL_NETWORKS	*	*	none		R03 - Management PC to Internal Networks	
	✓ 0/0 B	IPv4 TCP	192.168.199.71	*	*	WEB_PORTS	*	none		R04 - Management Web Access	
	✓ 0/0 B	IPv4 *	SITEA_MGMT subnets	*	*	*	*	none		Default allow LAN to any rule	
	✓ 0/0 B	IPv6 *	SITEA_MGMT subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

Figure 4a: Site A Management VLAN firewall rules allowing authorised management access.

Rules (Drag to Change Order)											
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
	✗ 0/924 B	IPv4 *	SITEA_EMPLOYEE subnets	*	192.168.199.0/24	*	*	none		block employee to management	
	✓ 0/1.24 MiB	IPv4 TCP/UDP	SITEA_EMPLOYEE subnets	*	This Firewall (self)	53 (DNS)	*	none		allow employee dns	
	✓ 0/0 B	IPv4 TCP	SITEA_EMPLOYEE subnets	*	192.168.130.0/24	PRINT_PORTS	*	none		Printer access for employee	
	✓ 0/0 B	IPv4 ICMP any	SITEA_EMPLOYEE subnets	*	This Firewall (self)	*	*	none		Employee to gateway	
	✓ 0/0 B	IPv4 ICMP any	SITEA_EMPLOYEE subnets	*	192.168.10.0/24	*	*	none		Allow site a employee icmp to hq	
	✓ 0/0 B	IPv4 TCP	SITEA_EMPLOYEE subnets	*	*	WEB_PORTS	*	none		allow employees web access	

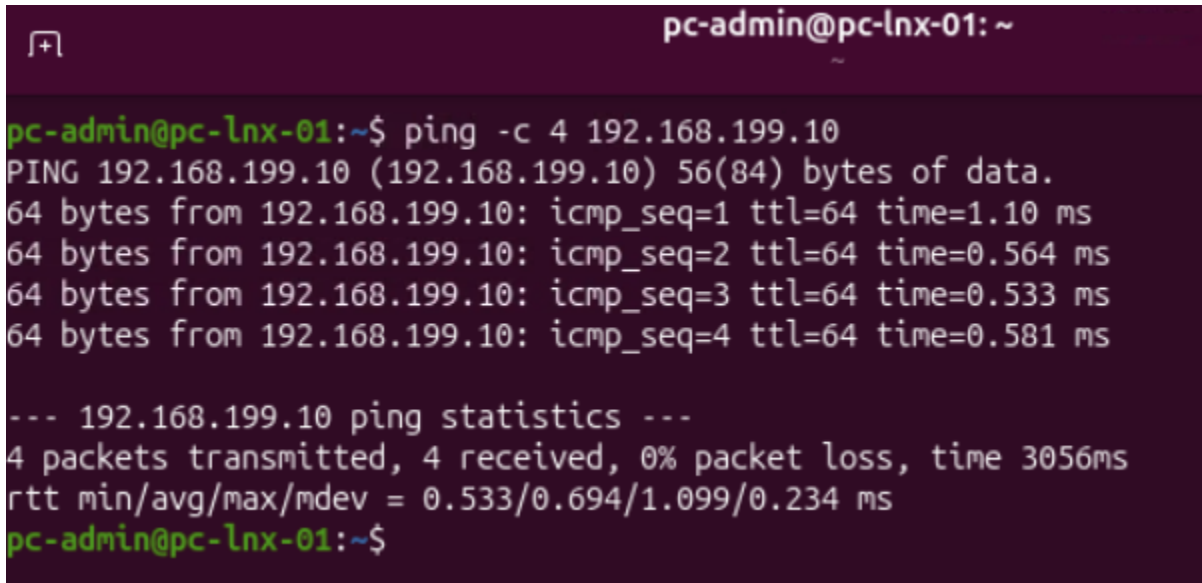
Figure 4b: Site A Employee VLAN rule blocking access to the Management VLAN.

Rules (Drag to Change Order)											
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
	✗ 0/0 B	IPv4 *	SITEA_EMP_WIFI subnets	*	192.168.199.0/24	*	*	none		Block employee wifi to management	
	✓ 0/0 B	IPv4 TCP/UDP	SITEA_EMP_WIFI subnets	*	This Firewall (self)	53 (DNS)	*	none		allow employee wifi dns	
	✓ 0/0 B	IPv4 TCP	SITEA_EMP_WIFI subnets	*	192.168.130.0/24	PRINT_PORTS	*	none		allow printer access	
	✓ 0/0 B	IPv4 ICMP any	SITEA_EMP_WIFI subnets	*	This Firewall (self)	*	*	none		Allow Employee WIFI ping to gateway	
	✓ 0/0 B	IPv4 TCP	SITEA_EMP_WIFI subnets	*	*	WEB_PORTS	*	none		allow employee web access	

Figure 4c: Site A Employee Wi-Fi rule blocking access to the Management VLAN.

5.3 Site A Switch Management

PCM-SITEA-SW01 was assigned the management IP address 192.168.199.10 on VLAN 99. While testing, SITE-A-MGMT-PC was able to connect to the switch successfully, confirming that the switch can be managed through VLAN 99 while keeping normal Employee and Visitor networks separate.

A terminal window with a dark background and light-colored text. The title bar at the top reads 'pc-admin@pc-lnx-01: ~'. The terminal shows a command 'ping -c 4 192.168.199.10' being executed. The output shows four successful ping responses, each with a TTL of 64 and a time between 0.533 ms and 1.10 ms. Below the individual responses, a summary line shows '4 packets transmitted, 4 received, 0% packet loss, time 3056ms' and 'rtt min/avg/max/mdev = 0.533/0.694/1.099/0.234 ms'. The prompt 'pc-admin@pc-lnx-01:~\$' is visible at the bottom.

```
pc-admin@pc-lnx-01:~$ ping -c 4 192.168.199.10
PING 192.168.199.10 (192.168.199.10) 56(84) bytes of data.
64 bytes from 192.168.199.10: icmp_seq=1 ttl=64 time=1.10 ms
64 bytes from 192.168.199.10: icmp_seq=2 ttl=64 time=0.564 ms
64 bytes from 192.168.199.10: icmp_seq=3 ttl=64 time=0.533 ms
64 bytes from 192.168.199.10: icmp_seq=4 ttl=64 time=0.581 ms

--- 192.168.199.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3056ms
rtt min/avg/max/mdev = 0.533/0.694/1.099/0.234 ms
pc-admin@pc-lnx-01:~$
```

Figure 5: Successful ping from SITE-A-MGMT-PC to the Site A switch management address 192.168.199.10.

6. Testing and Validation

Test	Expected Result	Actual Result	Status
HQ-MGMT-PC - HQ-FW	Management access permitted	HQ-MGMT-PC successfully accessed pfSense through VLAN 99.	PASS
HQ-MGMT-PC - HQ-SW-OVS	Switch management reachable	192.168.99.60 was reachable from HQ-MGMT-PC.	PASS
HQ Employee - HQ Management	Blocked and logged	Employee traffic to Management VLAN was blocked.	PASS
HQ Employee Wi-Fi - HQ Management	Blocked and logged	Employee Wi-Fi traffic to Management VLAN was blocked.	PASS
HQ-MGMT-PC - Grafana	TCP 3000 permitted	Grafana loaded from HQ-MGMT-PC	PASS
SITE-A-MGMT-PC - SITE-A-FW	Management access permitted	192.168.199.1 was reachable.	PASS
SITE-A-MGMT-PC - PCM-SITEA-SW01	Switch management reachable	192.168.199.10 was reachable.	PASS
SITE-A-MGMT-PC - Site A employee endpoint	Authorised internal reachability	Connection test succeeded.	PASS

The testing demonstrates that the approved management PCs could access the devices and services they need, while Employee and Employee Wi-Fi devices were still blocked from the Management VLAN. This showcases VLAN 99 works as intended, as it keeps management access separate at both locations.

```

Recent pfSense Events
: UNK 2026-10-05T06:00:31+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,54791,0,DF,6,tcp,60,192.168.199.30,8.8.8.8,58100,53,0,S,3216065697,,64240
: UNK 2026-10-05T06:00:34+00:00 _gateway filterlog[48615]: 1,6,,1000000103,vtnet2,match,block,in,4,0x0,,64,0,0,none,17,udp,312,0.0.0.0,255.255.255.255,68,67,292
: UNK 2026-10-05T06:00:34+00:00 _gateway filterlog[48615]: 1,129,,1787748345,vtnet2.10,match,block,in,4,0x0,,64,20465,0,DF,1,icmp,84,192.168.10.72,192.168.99.1,request,50760,364
: UNK 2026-10-05T06:00:30+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,54790,0,DF,6,tcp,60,192.168.199.30,8.8.8.8,58100,53,0,S,3216065697,,64240
  
```

Figure 6a: Grafana dashboard showing ICMP traffic from 192.168.10.72 being blocked from accessing the VLAN 99 gateway at 192.168.99.1.

```
Recent pfSense Events
: UNK 2026-10-05T06:13:22+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,32274,0,DF,6,tcp,60,192.168.199.71,8.8.8.8,48564,53,0,S,710767904,,64240,
: UNK 2026-10-05T06:13:21+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,32273,0,DF,6,tcp,60,192.168.199.71,8.8.8.8,48564,53,0,S,710767904,,64240,
: UNK 2026-10-05T06:13:20+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,32272,0,DF,6,tcp,60,192.168.199.71,8.8.8.8,48564,53,0,S,710767904,,64240,
: UNK 2026-10-05T06:13:20+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,10886,0,DF,6,tcp,60,192.168.199.20,8.8.8.8,35642,53,0,S,4168388549,,64240,
: UNK 2026-10-05T06:13:23+00:00 _gateway filterlog[48615]: 1,171,,1787829015,vtnet2.11,match,block,in,4,0x0,,64,5863,0,DF,1,icmp,84,192.168.11.72,192.168.99.1,request,60722,464
: UNK 2026-10-05T06:13:23+00:00 _gateway filterlog[48615]: 1,6,,1000000103,vtnet2,match,block,in,4,0x0,,64,0,0,none,17,udp,312,0.0.0.0,255.255.255.255,68,67,292
```

Figure 6b: Grafana dashboard showing ICMP traffic from the HQ Employee Wi-Fi host 192.168.11.72 being blocked from accessing the VLAN 99 gateway at 192.168.99.1.

7. Limitations and Future Improvements

- The prototype uses dedicated management PCs and IP-based firewall rules to control access. In a real network, stronger administrator authentication and multi-factor authentication could also be added.
- The final Site A switch management configuration should be included in the evidence if it is available.
- The management firewall rules should be reviewed over time and limited to the specific devices and services that actually need access.
- More devices can be added to VLAN 99 as the network grows, but they should still only be given the access they need instead of automatically trusting every device on the Management VLAN.

8. Conclusion

The VLAN 99 Management network was set up successfully at Brisbane HQ and Mine Site A to keep management access separate from user networks. As shown in testing, the approved management devices could access the required firewall, switch and monitoring services, while Employee and Employee Wi-Fi devices were blocked from accessing the Management VLAN. The blocked traffic was also visible in the centralised Grafana dashboard, confirming that the management restrictions were working as intended.